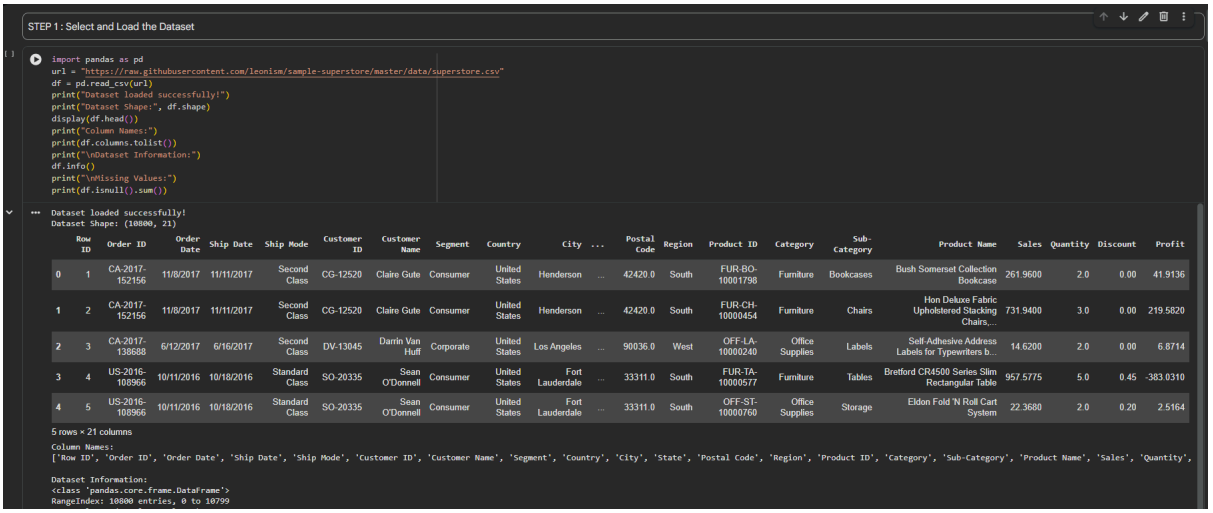
LAB SHEET 4

Experiment Name: Advanced Data Visualization using Matplotlib and Seaborn Steps

1. Select a real-world dataset such as the Superstore Sales Dataset, Netflix Dataset, COVID-19 Dataset, or World Happiness Report.
2.bImport the required libraries (Pandas, Matplotlib, and Seaborn) and load the cleaned dataset.
3. Create bar charts to compare categorical variables such as sales by region, genre distribution, or product categories.
4. Generate line charts to analyze trends over time, such as monthly sales, yearly movie releases, or daily COVID-19 cases.
5. Plot histograms to study the distribution of numerical variables such as sales, ratings, or customer age.
6. Create box plots to identify outliers and compare distributions across different categories.
7. Generate scatter plots to analyze relationships between two numerical variables, such as advertising expenditure vs. sales or budget vs. revenue.
8. Create a correlation heatmap using Seaborn to visualize relationships among numerical features.
9. Customize the visualizations by adding titles, axis labels, legends, annotations, and appropriate color palettes to improve readability.
10. Interpret each visualization and prepare a summary highlighting key business insights obtained from the graphical analysis.

Solution ->



STEP 1 : Select and Load the Dataset

```
import pandas as pd
url = "https://raw.githubusercontent.com/leonism/sample-superstore/master/data/superstore.csv"
df = pd.read_csv(url)
print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)
display(df.head())
print("Column Names:")
print(df.columns.tolist())
print("\nDataset Information:")
df.info()
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset loaded successfully!
Dataset Shape: (10000, 21)
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
0	1	CA-2017-152156	11/8/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420.0	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600	2.0	0.00	41.9136
1	2	CA-2017-152156	11/8/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420.0	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3.0	0.00	219.5820
2	3	CA-2017-138688	6/12/2017	6/16/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036.0	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200	2.0	0.00	6.8714
3	4	US-2016-108966	10/11/2016	10/18/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311.0	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5.0	0.45	-383.0310
4	5	US-2016-108966	10/11/2016	10/18/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311.0	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680	2.0	0.20	2.5164

5 rows × 21 columns

```
Column Names:
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales', 'Quantity',

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 10799
```

```
5 rows × 21 columns
Column Names:
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales', 'Quantity',

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10800 entries, 0 to 10799
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         10800 non-null  object
 1   Order ID       10800 non-null  object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9983 non-null   float64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   float64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(5), object(16)
memory usage: 1.7+ MB
```

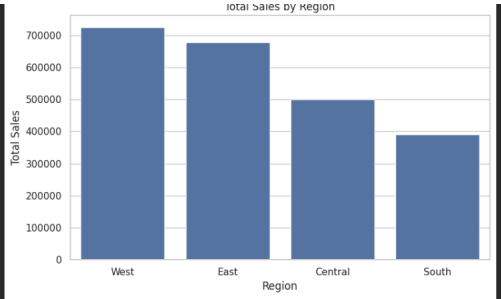
STEP 2 : Import Required Libraries

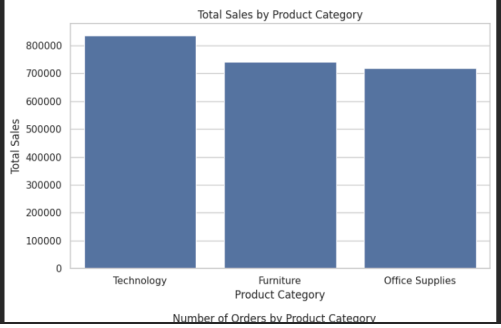
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
print("All required libraries imported successfully!")
```

```
All required libraries imported successfully!
```

STEP 3 : Create Bar Charts

```
region_sales = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
sns.barplot(
    x=region_sales.index,
    y=region_sales.values
)
plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
category_sales = df.groupby('Category')['Sales'].sum().sort_values(
    ascending=False
)
plt.figure(figsize=(8, 5))
sns.barplot(
    x=category_sales.index,
    y=category_sales.values
)
plt.title("Total Sales by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
category_count = df['Category'].value_counts()
plt.figure(figsize=(8, 5))
sns.barplot(
    x=category_count.index,
    y=category_count.values
)
plt.title("Number of Orders by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Number of Orders")
plt.tight_layout()
plt.show()
```

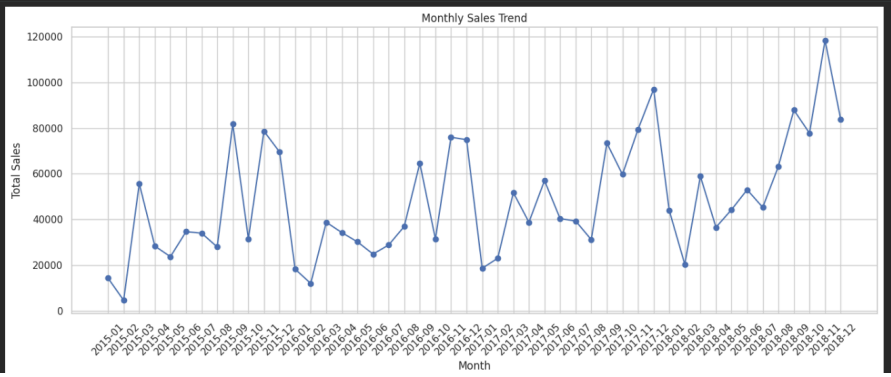




Number of Orders by Product Category

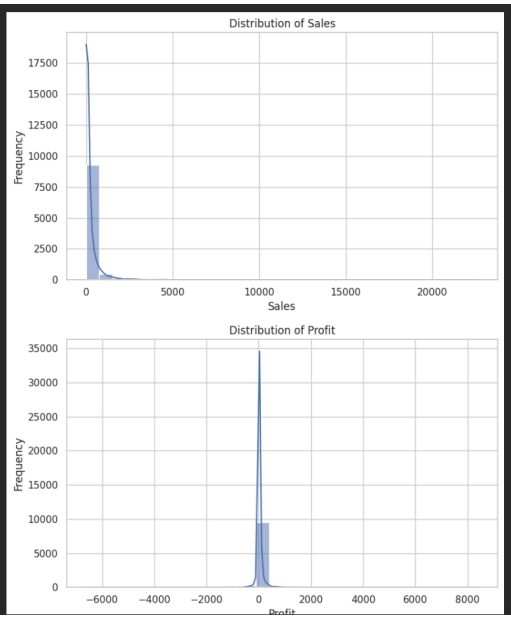
STEP 4 : Generate Line Charts

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['YearMonth'] = df['Order Date'].dt.to_period('M')
monthly_sales = df.groupby('YearMonth')['Sales'].sum()
monthly_sales.index = monthly_sales.index.astype(str)
plt.figure(figsize=(14, 6))
plt.plot(
    monthly_sales.index,
    monthly_sales.values,
    marker='o'
)
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



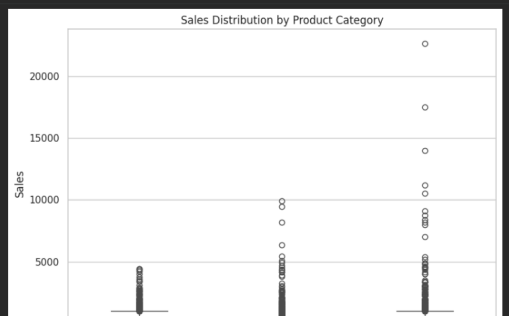
STEP 5 : Plot Histograms

```
plt.figure(figsize=(8, 5))
sns.histplot(
    df['Sales'],
    bins=30,
    kde=True
)
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
sns.histplot(
    df['Profit'],
    bins=30,
    kde=True
)
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
sns.histplot(
    df['Discount'],
    bins=20,
    kde=True
)
plt.title("Distribution of Discount")
plt.xlabel("Discount")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



STEP 6 : Create Box Plots

```
plt.figure(figsize=(8, 6))
sns.boxplot(
    x='Category',
    y='Sales',
    data=df
)
plt.title("Sales Distribution by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 6))
sns.boxplot(
    x='Region',
    y='Profit',
    data=df
)
plt.title("Profit Distribution by Region")
plt.xlabel("Region")
plt.ylabel("Profit")
plt.tight_layout()
plt.show()
```



STEP 7 : Generate Scatter Plots

```
plt.figure(figsize=(9, 6))
sns.scatterplot(
    x='Sales',
    y='Profit',
    data=df,
    alpha=0.6
)
plt.title("Relationship Between Sales and Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.tight_layout()
plt.show()
```

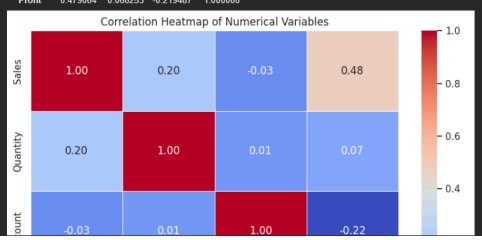


STEP 8 : Create Correlation Heatmap

```
numerical_columns = [
    'Sales',
    'Quantity',
    'Discount',
    'Profit'
]
correlation_matrix = df[numerical_columns].corr()
print("Correlation Matrix:")
display(correlation_matrix)
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)
plt.title("Correlation Heatmap of Numerical Variables")
plt.tight_layout()
plt.show()
```

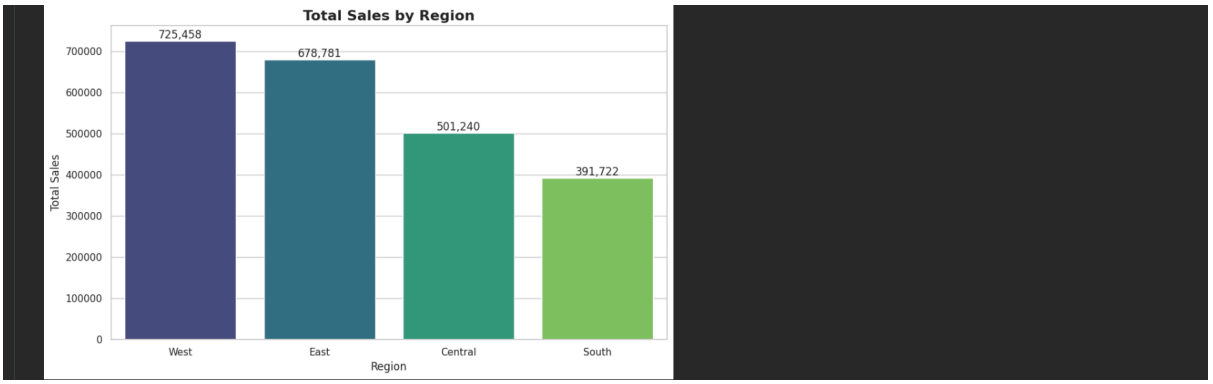
```
Correlation Matrix:
```

	Sales	Quantity	Discount	Profit
Sales	1.000000	0.200795	-0.028190	0.479064
Quantity	0.200795	1.000000	0.008623	0.066253
Discount	-0.028190	0.008623	1.000000	-0.219487
Profit	0.479064	0.066253	-0.219487	1.000000



STEP 9 : Customize Visualizations

```
[7]
plt.figure(figsize=(10, 6))
ax = sns.barplot(
    x=region_sales.index,
    y=region_sales.values,
    palette="viridis",
    hue=region_sales.index,
    legend=False
)
plt.title(
    "Total Sales by Region",
    fontsize=16,
    fontweight='bold'
)
plt.xlabel("Region")
plt.ylabel("Total Sales")
for i, value in enumerate(region_sales.values):
    ax.text(
        i,
        value,
        f"{value:,.0f}",
        ha='center',
        va='bottom'
    )
plt.tight_layout()
plt.show()
```



Questions

1. Why is data visualization considered an essential component of data analytics?
2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?
3. Which type of chart would you choose to compare sales across different regions? Justify your answer.
4. What information can be obtained from a histogram and a box plot?
5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?
6. What is a correlation heatmap? How can it assist in feature selection for machine learning?
7. Why is chart customization (titles, labels, legends, colors) important in data visualization?
8. What factors should be considered while selecting an appropriate visualization for a dataset?
9. How can misleading visualizations affect business decision-making? Give a real-world example.
10. Explain how data visualization supports storytelling and business intelligence in organizations.

Answers -> **1. Why is data visualization considered an essential component of data analytics?**

Data visualization is the process of representing data using **charts, graphs, plots, and diagrams**. It is an essential part of data analytics because large amounts of numerical data can be difficult to understand in tabular form.

It helps analysts to:

- Identify **trends and patterns** quickly.
- Detect **outliers and unusual values**.
- Compare different categories or groups.
- Understand relationships between variables.
- Communicate analytical results effectively.
- Support better and faster **business decision-making**.

Example: A line chart of monthly sales can immediately show whether sales are increasing or decreasing.

2.

Matplotlib	**Seaborn**
General-purpose visualization library	Statistical visualization library built on Matplotlib
Provides detailed control over plots	Provides simpler and more attractive statistical plots
Requires more code for some complex visualizations	Requires less code for statistical graphs
Suitable for basic and highly customized plots	Suitable for distributions, correlations, categorical data, etc.
Works directly with many data structures	Works particularly well with Pandas DataFrames

3. A **bar chart** is the most appropriate choice for comparing sales across different regions.

For example:

- Delhi → ₹50,000
- Mumbai → ₹75,000
- Chennai → ₹60,000
- Kolkata → ₹45,000

A bar chart makes it easy to compare the size of sales values between regions.

Why use a bar chart?

- It is simple and easy to understand.
- It clearly compares different categories.
- Differences between regions can be identified quickly.
- Both vertical and horizontal bar charts can be used.

Example: A horizontal bar chart is particularly useful when there are many regions or long region names.

4. Histogram

A histogram shows the **distribution of numerical data** by dividing values into intervals called bins.

It helps identify:

- Distribution shape.
- Frequency of observations.
- Central tendency.
- Spread of data.
- Skewness.
- Possible multiple peaks.

For example, a histogram of student marks can show whether most students scored between 60–70 marks.

Box Plot

A box plot summarizes data using the **five-number summary**:

1. Minimum
2. First quartile (Q1)
3. Median
4. Third quartile (Q3)
5. Maximum

It helps identify:

- Median.
- Interquartile range (IQR).

- Data spread.
- Possible outliers.
- Skewness.
- Differences between groups.

5. A **scatter plot** represents the relationship between two numerical variables using individual points.

For example, suppose we want to study the relationship between:

- Study hours
- Examination marks

Each point represents one student's study hours and marks.

A scatter plot helps determine whether variables have:

- **Positive relationship:** Both variables tend to increase together.
- **Negative relationship:** One variable increases while the other decreases.
- **No relationship:** There is no clear pattern.
- **Non-linear relationship:** Variables have a curved or other non-linear pattern.

6. A **correlation heatmap** is a graphical representation of the correlation coefficients between multiple variables. Different colors or shades represent different correlation strengths.

The **Pearson correlation coefficient** generally ranges from:

-1 to +1

- **+1** → Perfect positive correlation
- **0** → No linear correlation
- **-1** → Perfect negative correlation

Use in feature selection

A correlation heatmap can help identify:

- Features strongly correlated with the **target variable**.
- Features that are highly correlated with each other.
- **Multicollinearity** among independent variables.

If two input features have very high correlation, one of them may sometimes be removed to reduce redundancy.

7. Chart customization improves the **clarity, readability, and interpretation** of a visualization.

Important elements include:

- **Title:** Explains what the chart represents.
- **Axis labels:** Explain what each axis measures.
- **Legend:** Identifies different categories or data series.
- **Colors:** Help distinguish between categories.
- **Annotations:** Highlight important values or observations.
- **Gridlines:** Can make numerical comparisons easier.

8. The following factors should be considered:

1. **Type of data**
   Determine whether the data is numerical, categorical, time-series, or geographical.
2. **Purpose of analysis**
   Decide whether you want to compare, show trends, study relationships, or show distributions.
3. **Number of variables**
   Different charts are suitable for one, two, or multiple variables.
4. **Number of categories**
   A pie chart may work for a few categories, while a bar chart is usually better for many categories.
5. **Audience**
   The visualization should match the technical knowledge of the audience.
6. **Data size**
   Very large datasets may require specialized visualization techniques.
7. **Readability**
   Avoid unnecessary complexity and excessive colors.
8. **Accuracy**
   The visualization should represent the data honestly and avoid misleading scales.

Examples:

- Comparison → **Bar chart**

- Trend over time → **Line chart**
- Distribution → **Histogram**
- Relationship → **Scatter plot**
- Correlation → **Heatmap**
- Outliers → **Box plot**

9. Misleading visualizations can cause managers and organizations to **misinterpret data and make incorrect decisions**.

Common causes include:

- Manipulating the axis scale.
- Starting a bar chart's y-axis at an inappropriate value.
- Using misleading proportions.
- Showing only selected data.
- Using confusing colors.
- Removing important context.

Example

Suppose a company's sales increased from **₹100 lakh to ₹105 lakh**.

The actual increase is only **5%**. However, if a graph starts its y-axis at ₹99 lakh instead of ₹0, the visual difference between the two bars may appear extremely large.

Management might incorrectly conclude that sales have increased dramatically and invest heavily in expansion.

Therefore, visualizations should present data **accurately, objectively, and with appropriate context**.

10. Data visualization plays an important role in **data storytelling and Business Intelligence (BI)**.

It converts raw data into understandable visual information that helps organizations identify what happened, why it happened, and what actions should be taken.

Role in data storytelling

A typical data story can follow this structure:

Data → Visualization → Insight → Decision → Action

For example:

1. A company collects monthly sales data.
2. A line chart shows that sales have decreased for three consecutive months.
3. A regional bar chart shows that the decline is mainly from one region.
4. Further analysis identifies reduced customer demand.
5. Management develops a new marketing strategy for that region.

Role in Business Intelligence

Visualization helps organizations:

- Monitor **KPIs** and performance.
- Track sales and revenue.
- Identify market trends.
- Compare departments or regions.
- Monitor customer behavior.
- Detect problems and anomalies.
- Support strategic decision-making.
- Present information through interactive dashboards.